

National Institute of Technology, Kurukshetra
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Sub:-Power System Analysis, Mid Sem-II
B.Tech 2nd year, Section-E-4 to E-6

Date:- 07-04-2026
Max. Marks:-15

Time:- 50 Minutes

Q.1: The currents in a 3-phase unbalanced system are :

$$I_R = (12 + j 6) \text{ A} ; I_Y = (12 - j 12) \text{ A} ; I_B = (-15 + j 10) \text{ A}$$

The phase sequence is RYB. Calculate the zero, positive and negative sequence components of the currents in all phases.

Q.2: Derive the expression of fault current for a LLG fault on generator terminal and draw the resultant sequence network.

Q.3: Derive the expression of bus admittance matrix using singular transformation method and determine the per-unit bus admittance matrix of the following network.

